

Homework 2: Conditional Expectation and Bayes Prediction

Goriola-Obafemi Babatunde Sukanmi

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Setup

Let $X \sim \text{Unif}[-1, 1]$ and $Z \sim \text{Exp}(1)$ be independent, with

$$f_Z(z) = e^{-z}, \quad z \geq 0,$$

and let $Y = X + Z$.

1. Conditional density of Y given X

Given $X = x$, we have $Y = x + Z$, so $Y \mid X = x$ is simply Z shifted by x . Since $Z \geq 0$, it follows that $Y \geq x$, and the density is

$$f_{Y|X}(y \mid x) = f_Z(y - x) = e^{-(y-x)}, \quad y \geq x,$$

and $f_{Y|X}(y \mid x) = 0$ for $y < x$.

2. Conditional mean and median

Since $E[Z] = 1$ for $Z \sim \text{Exp}(1)$,

$$E[Y \mid X = x] = x + E[Z] = x + 1.$$

For the median $m(x)$, we solve $P(Z \leq m(x) - x) = \frac{1}{2}$, i.e.

$$1 - e^{-(m(x)-x)} = \frac{1}{2} \implies m(x) - x = \ln 2 \implies m(x) = x + \ln 2.$$

3. Bayes predictors

(a) Squared loss

The Bayes predictor under squared loss is the conditional mean:

$$f^*(x) = E[Y \mid X = x] = x + 1.$$

(b) Absolute loss

The Bayes predictor under absolute loss is the conditional median:

$$f^*(x) = \text{median}(Y \mid X = x) = x + \ln 2.$$

4(a). The family $f_a(x) = x + a$, $0 \leq a \leq 2.5$

Since $Y - f_a(X) = (X + Z) - (X + a) = Z - a$, the residual does not depend on X at all, it depends only on Z and a .

Squared risk

$$R_{sq}(a) = E[(Z - a)^2] = \text{Var}(Z) + (E[Z] - a)^2 = 1 + (1 - a)^2 = a^2 - 2a + 2.$$

Differentiating and setting to zero, $2a - 2 = 0 \implies a = 1$, giving

$$R_{sq}(1) = 1.$$

This matches the Bayes predictor from part 3(a): $f_1(x) = x + 1$.

Absolute risk

Splitting the integral at $z = a$:

$$R_{abs}(a) = E|Z - a| = \int_0^a (a - z)e^{-z} dz + \int_a^\infty (z - a)e^{-z} dz = a - 1 + 2e^{-a}, \quad a \geq 0.$$

Differentiating, $1 - 2e^{-a} = 0 \implies a = \ln 2$, giving

$$R_{abs}(\ln 2) = \ln 2.$$

This matches the Bayes predictor from part 3(b): $f_{\ln 2}(x) = x + \ln 2$.

Why this is not a coincidence

The family $f_a(x) = x + a$ was chosen with slope exactly 1 on x , matching the slope of both true Bayes predictors. Minimizing over a within this family therefore recovers the unconstrained Bayes predictor exactly, rather than only approximating it, because the only missing piece was the right additive constant.